

ENERGY OBSERVABILITY ACROSS FOURIER–GALERKIN RESOLUTIONS OF THREE-DIMENSIONAL NAVIER–STOKES

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Consider the unforced periodic three-dimensional incompressible Navier–Stokes equation after finite cubic Fourier–Galerkin truncation. This note separates two questions that are often conflated: how many simultaneous energy measurements are retained, and how deeply their time evolution must be observed before they become locally complete modulo symmetry. For every finite cubic resolution we derive sharp structural ceilings for one scalar total-energy reader and for the full shell-energy reader, and show that the resulting jet ranks are independent of every positive viscosity. Exact modular certificates attain the ceilings at the earliest structurally possible orders in five tested reader–resolution cases: scalar energy at $N = 1, 2$ and shell energy at $N = 1, 2, 3$. In particular, the shell reader now saturates at the structural minimum for three consecutive finite cutoffs. The finite results motivate a resolution-indexed saturation conjecture and an explicit measurement–time trade-off. We also state the precise finite-dimensional bridge from rank saturation to a quantitative local inverse certificate; that bridge requires additional conditioning and branch bounds and is not supplied by rank alone. No continuum regularity claim is made.

1. Resolution, readers, and all- N ceilings

For $N \geq 1$, retain

$$(1) \quad K_N = \{-N, \dots, N\}^3 \setminus \{0\},$$

with real phase-space dimension

$$(2) \quad \boxed{d_N = 2((2N + 1)^3 - 1)}.$$

Write

$$(3) \quad \dot{x} = F_\nu(x) = \nu Ax + B(x, x), \quad \nu > 0,$$

where B is the projected quadratic Navier–Stokes interaction. Let

$$(4) \quad Q_N = \{|k|^2 : k \in K_N\}, \quad m_N = |Q_N|,$$

and let $I = (I_q)_{q \in Q_N} \in \mathbb{R}^{m_N}$ denote the exact shell energies. Their sum is total retained kinetic energy

$$(5) \quad E = \mathbf{1}^T I.$$

Define the scalar and shell Lie jets

$$(6) \quad \mathcal{E}_R = (E, \mathcal{L}_F E, \dots, \mathcal{L}_F^R E),$$

$$(7) \quad \mathcal{J}_R = (I, \mathcal{L}_F I, \dots, \mathcal{L}_F^R I).$$

Translation by $a \in \mathbb{T}^3$ acts as $(T_a \widehat{u})_k = e^{ik \cdot a} \widehat{u}_k$. The dynamics is translation equivariant and both readers are invariant. At generic states the translation orbit is three-dimensional, so both jet ranks are at most $d_N - 3$.

The scalar jet has $R + 1$ coordinates. For the shell reader, the exact shell balance is

$$(8) \quad J_1 + L_\nu J_0 = \eta^{NS}, \quad \mathbf{1}^T \eta^{NS} = 0,$$

where L_ν is the fixed shellwise viscous operator. Repeated differentiation yields one scalar dependence in each new m_N -component shell block. Hence:

Theorem 1 (Universal finite-resolution ceilings). *For every $N \geq 1$, every $\nu > 0$, and every $R \geq 0$,*

$$(9) \quad \boxed{\text{rank } D\mathcal{E}_R \leq \min(R + 1, d_N - 3),}$$

$$(10) \quad \boxed{\text{rank } D\mathcal{J}_R \leq \min(m_N + (m_N - 1)R, d_N - 3).}$$

Therefore the symmetry ceiling cannot be reached before

$$(11) \quad \boxed{R_E^{\min} = d_N - 4,}$$

$$(12) \quad \boxed{R_I^{\min} = \left\lceil \frac{d_N - 3 - m_N}{m_N - 1} \right\rceil.}$$

2. Positive-viscosity universality

The viscosity parameter does not generate a new rank problem at each positive value.

Theorem 2 (Positive-viscosity rank universality). *Fix N and either energy reader $h \in \{E, I_q : q \in Q_N\}$. For every $r \geq 0$ and $\nu > 0$,*

$$(13) \quad \mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+2} \mathcal{L}_{F_1}^r h(y),$$

$$(14) \quad D_x \mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+1} D_y \mathcal{L}_{F_1}^r h(y).$$

Consequently every scalar- and shell-energy jet rank is independent of the value of $\nu > 0$.

Proof. Set $x(t) = \nu y(s)$ and $s = \nu t$. Since B is quadratic, $F_\nu(\nu y) = \nu^2 F_1(y)$. Each energy reader is quadratic, giving the displayed identities. For $\nu > 0$ the induced state and row scalings are invertible, so rank is unchanged. $\square$

The endpoint $\nu = 0$ is different. In particular total kinetic energy is conserved for the finite Euler system, so the scalar energy jet collapses to $(E, 0, 0, \dots)$.

3. Exact saturation certificates

The certificates propagate formal Taylor coefficients of the Galerkin flow and tangent system. For polynomial F and reader h ,

$$(15) \quad h(\Phi_t x) = \sum_{r \geq 0} \frac{t^r}{r!} \mathcal{L}_F^r h(x).$$

Thus over a prime field $\mathbb{F}_p$ with $p > R$, Taylor-block and Lie-jet Jacobians have identical rank through order R . A nonzero maximal minor modulo a good prime implies that the corresponding characteristic-zero rational minor polynomial is not identically zero; the universal ceilings then give equality of the generic characteristic-zero rank.

The exact certificates currently give:

$$(16) \quad \begin{array}{c|c|c|c|c|c} N & d_N & \text{reader} & m & R_{\min} & \text{certified rank} \\ \hline 1 & 52 & \text{total energy} & 1 & 48 & 49 \\ 1 & 52 & \text{shell energies} & 3 & 23 & 49 \\ 2 & 248 & \text{total energy} & 1 & 244 & 245 \\ 2 & 248 & \text{shell energies} & 9 & 30 & 245 \\ 3 & 684 & \text{shell energies} & 18 & 39 & 681 \end{array}$$

Every row reaches the translation ceiling $d_N - 3$ at the earliest order allowed by the corresponding structural bound.

For $N = 2$, the exact total-energy certificate over $\mathbb{F}_{251}$ satisfies

$$(17) \quad \boxed{\text{rank}_{\mathbb{F}_{251}} D\mathcal{E}_{244} = 245.}$$

The nine shell values are

$$(18) \quad Q_2 = \{1, 2, 3, 4, 5, 6, 8, 9, 12\},$$

and the exact shell calculation over the same good prime gives the new middle-resolution certificate

$$(19) \quad \boxed{\text{rank}_{\mathbb{F}_{251}} D\mathcal{J}_{30} = 245,}$$

while at the preceding order

$$(20) \quad \boxed{\text{rank}_{\mathbb{F}_{251}} D\mathcal{J}_{29} = 241 < 245.}$$

Hence $R = 30$ is both attained and structurally earliest. The recorded $N = 2$ shell milestones are

$$(21) \quad \begin{array}{c|cccccccc} R & 0 & 1 & 5 & 10 & 20 & 25 & 29 & 30 \\ \hline \text{rank} & 9 & 17 & 49 & 89 & 169 & 209 & 241 & 245, \end{array}$$

which equal $\min(9 + 8R, 245)$ at every recorded order.

For $N = 3$, the shell set is

$$(22) \quad Q_3 = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 17, 18, 19, 22, 27\},$$

so $m_3 = 18$, and an exact projected calculation over $\mathbb{F}_{683}$ gives

$$(23) \quad \boxed{\text{rank}_{\mathbb{F}_{683}} D\mathcal{J}_{39} = 681.}$$

The recorded $N = 3$ milestones are

$$(24) \quad \begin{array}{c|cccccccc} R & 0 & 5 & 10 & 20 & 30 & 35 & 38 & 39 \\ \hline \text{rank} & 18 & 103 & 188 & 358 & 528 & 613 & 664 & 681, \end{array}$$

which follow $\min(18 + 17R, 681)$ at every recorded order.

Proposition 1 (Certified finite saturation). *At the five reader–resolution pairs in the table, and for every $\nu > 0$, the corresponding energy jet has generic local rank $d_N - 3$ at its earliest structurally possible order.*

In particular, the shell reader is now exactly certified at three consecutive finite cutoffs,

$$(25) \quad \boxed{(N, R_I^{\min}, d_N - 3) = (1, 23, 49), (2, 30, 245), (3, 39, 681).}$$

This is finite evidence, not an induction theorem and not a proof for arbitrary finite N .

At generic states the translation action is free; for example, nonzero coefficients at the three axial modes force a fixing translation to satisfy $e^{ia_1} = e^{ia_2} = e^{ia_3} = 1$. Therefore rank $d_N - 3$ means local completeness modulo continuous translation: the kernel of the saturated jet differential equals the tangent space of the translation orbit. This is local, not global, injectivity.

4. Resolution-indexed conjecture and measurement–time trade-off

The exact cases suggest that the structural ceilings may be generically sharp at every finite resolution.

Conjecture 1 (Optimal energy-readout saturation). *For every finite cubic truncation K_N and every $\nu > 0$,*

$$(26) \quad \boxed{\text{rank}_{\text{gen}} D\mathcal{E}_{d_N-4} = d_N - 3,}$$

and

$$(27) \quad \boxed{\text{rank}_{\text{gen}} D\mathcal{J}_{R_I^{\min}} = d_N - 3.}$$

Only the certified rows above are proved; the conjecture is not a statement about $N \rightarrow \infty$ or continuum Navier–Stokes. In the finite-first reading, the desired general result is a construction or proof schema that accepts any particular finite N and certifies the corresponding finite minor. The three consecutive shell cases do not by themselves supply such a schema.

Nevertheless, the universal ceilings already expose an exact design trade-off. A scalar reader requires at least

$$(28) \quad R_E^{\min} = d_N - 4 = 2(2N + 1)^3 - 6 = \Theta(N^3)$$

time derivatives before symmetry-limited completeness is even possible. By contrast, the m_N -channel shell reader has lower bound (12). At $N = 3$ this changes the minimum admissible temporal depth from

$$(29) \quad \boxed{1 \text{ channel : } R \geq 680}$$

to the exactly saturated

$$(30) \quad \boxed{18 \text{ channels} : R = 39.}$$

Thus the finite theory makes precise a measurement-time principle:

$$(31) \quad \boxed{\text{more simultaneous dynamical readout} \longleftrightarrow \text{less temporal observability depth.}}$$

This does not yet identify a physical sensor layout: shell energies are spectral observables, and stable recovery from noisy high-order derivatives remains open. It does identify the quantity that a practical sensor-design program should minimize: temporal depth subject to saturating the symmetry-limited observable state.

5. Bridge from observability to quantitative inversion

Rank saturation and quantitative reconstruction are deliberately separated. Fix one finite N and suppose a saturated observation jet has generic rank $d_N - 3$. At a point where the translation action is locally free, choose an explicit transverse symmetry slice S_N of dimension $d_N - 3$ and select a square set of observation coordinates $H_N : S_N \rightarrow \mathbb{R}^{d_N-3}$ with

$$(32) \quad \det DH_N(x_*) \neq 0.$$

This gives a local inverse in finite dimension, but not yet a numerical error radius.

For a declared convex branch $B_N \subset S_N$ and a fixed preconditioner A_N , define

$$(33) \quad q_N = \sup_{x \in B_N} \|I - A_N DH_N(x)\|.$$

If $q_N < 1$, then

$$(34) \quad \boxed{\|x - z\| \leq \frac{\|A_N\|}{1 - q_N} \|H_N(x) - H_N(z)\|,}$$

and bounded measurement plus forward-residual uncertainty yields a finite retained-state certificate

$$(35) \quad \boxed{\rho_N \leq \frac{\|A_N\|}{1 - q_N} (\sigma_{\text{meas}} + \sigma_{\text{res}}).}$$

This is the precise interface between the energy-observability programme and the EPSC quantitative-inversion programme. Saturation supplies the structural possibility of a square local chart; a branch-wide $q_N < 1$ enclosure supplies quantitative stability. Neither statement implies the other globally.

At $N = 1$, a separate exact characteristic-zero construction already supplies a 49-dimensional translation slice, a square shell-energy chart, an exact rational center inverse, and a centered entrywise enclosure on

$$(36) \quad \|x - x_*\|_\infty \leq 10^{-17}$$

with reproduced defect

$$(37) \quad \boxed{q_1 = 0.08058674502845 < 1/2.}$$

Thus the next finite quantitative target suggested by the new $N = 2$ shell certificate is not another rank computation: it is a 245-dimensional symmetry-fixed shell chart at $R = 30$ together with an explicit preconditioner and certified branch-wide defect $q_2 < 1$. Extending this package constructively to arbitrary finite N is a stronger problem than the saturation conjecture.

Any omitted-mode or continuum estimate belongs to a separate outer adapter. If a finite inner error ρ_N and an independently justified orthogonal tail error β_N are both available, they may subsequently be combined as $\sqrt{\rho_N^2 + \beta_N^2}$. No such outer statement is needed for the finite observability or finite inversion results in this note.

6. Scope and verification record

The results are finite-dimensional observability statements plus an explicit interface to a separate finite-dimensional quantitative-inverse criterion. They do not prove continuum regularity or singularity formation, global reconstruction, stable noisy differentiation, or computational speed-up. Their precise content is the combination of universal finite- N structural ceilings, positive-viscosity rank universality, exact ceiling saturation at the certified reader-resolution pairs above, and the logical bridge that identifies which extra certificate is required to turn structural observability into a finite error radius.

The derivation record, executable checkers, and machine-readable certificates are maintained at

<https://github.com/morrocwi/readout-problem-navier-stokes>.

The principal records include

`reproduction/NS_POSITIVE_VISCOSITY_OBSERVABILITY.md`,
`reproduction/checks/check_k2_energy_observability.py`,
`reproduction/checks/check_k2_shell_energy_observability.py`, and
`reproduction/checks/check_k3_shell_energy_observability.py`.